

Epistemic Uncertainty as a Driver of Performance in Wearable Monitoring of Parkinson’s Disease: A Real-World Proof-of-Concept Study

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Abstract—Wearable sensing systems can improve mobility management in people with Parkinson’s disease (PwP), yet real-world deployment remains limited due to uncertainty in sensor measurements and algorithmic predictions. Uncertainty can be aleatoric (e.g., measurement noise) or epistemic (e.g., limited system knowledge). Here, we investigated whether performance improvements in uncertainty-aware wearable sensing systems arise primarily from reductions in epistemic rather than aleatoric uncertainty. Using real-world inertial recordings from four PwP, we developed a personalized uncertainty-aware artificial intelligence pipeline combining time-series forecasting and convolutional neural networks to detect ON/OFF motor states. The system achieved a group-level balanced accuracy of 0.74. Ablation of uncertainty awareness reduced performance by 11%, indicating that uncertainty modulation enhances performance. Analysis of prediction windows showed that performance gains were predominantly associated with reductions in epistemic uncertainty. Findings suggest that modulating epistemic uncertainty may be key to improving wearable sensing systems for PwP and informing technical validation and regulatory approval.

I. INTRODUCTION

PRESERVING mobility is one of the highest priorities for people living with Parkinson’s disease (PD) [1]. By enabling objective and granular capture and assistance of motor behavior in daily life, wearable systems are promising to support treatment and enhance monitoring. Despite this promise, adoption of wearable systems in PD care remains limited. A major barrier involves handling the inherent variability in the sensing layer, which leads to ambiguity between mobility-related and unrelated features, especially in real-world settings [2]. Because sensing remains imperfect, the propagated signal inevitably carries distortion, leading to system-level uncertainty [3]. Recent work demonstrates that integrating *uncertainty awareness* can improve overall system performance [4]. Clarifying the factors underlying this performance improvement holds practical importance: first, handling uncertainty is a prerequisite for regulatory approval of wearable systems in PD [2], [3], [5]; second, it could inform design guidelines and technical validation strategies to mitigate the influence of these mechanisms on uncertainty in wearable systems for PD.

Uncertainty can be *aleatoric* or *epistemic*. *Aleatoric* uncertainty stems from inherent measurement noise, and *epistemic* uncertainty reflects limited knowledge within the system (e.g., abrupt changes in symptom manifestation). *Aleatoric* uncertainty is irreducible, but *epistemic* uncertainty can be reduced. One reason why *uncertainty awareness* may improve system performance is that it disproportionately reduces *epistemic* rather than *aleatoric* uncertainty. Here, we tested this hypothesis through a case study in which a personalized, uncertainty-aware, artificial intelligence–based wearable sensing system was designed for people with PD (PwP). The system estimates ON/OFF motor state from real-world gait. This case study was selected because it represents one of the areas in PD where the need for wearable monitoring is the most pressing (e.g., during medication titration or remote monitoring of motor fluctuations) [6], [7]. In this context, communicating uncertainty could facilitate broader adoption of wearable systems by enabling more transparent and trustworthy monitoring [3]. The objective was to determine whether system performance improvements under *uncertainty awareness* are primarily driven by reductions in *epistemic* rather than *aleatoric* uncertainty.

II. MATERIALS AND METHODS

A. Dataset

We present a secondary analysis of data from the Levodopa Response Study, details of which are reported elsewhere [8]. Data were obtained from four people with PD (PwP) experiencing motor fluctuations (25% female; median $\pm$ IQR age: 61 $\pm$ 8 years; MDS-UPDRS III ON: 37 $\pm$ 8; disease duration: 7 $\pm$ 4 years). All participants provided written informed consent, and the

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protocol was approved by relevant institutional review boards (#2014P000847). The dataset included participant demographics and clinical characteristics, two days of continuous lower-back inertial recordings during daily life (Shimmer3, Shimmer Research Ltd., 50 Hz), and time-stamped dopaminergic medication diaries.

B. Data Preprocessing

All participants underwent identical preprocessing. Gait sequences longer than 30 seconds were identified and segmented into continuous, nonoverlapping 5-second windows [9]. For each window, a signal magnitude vector a_{mag} was calculated. All a_{mag} were resampled to 100 Hz.

C. Personalized AI-Based Wearable System

The system integrates two sequential components. The first component applies a pre-trained time-series forecasting model to raw a_{mag} windows to predict signal evolution with a forecasting horizon of 50 ms [10]. In addition to predictions, the model generates confidence intervals (CIs) that quantify predictive uncertainty for each window. The median CI width per window was used as an uncertainty measure. The second component consists of a deep convolutional neural network (CNN) trained to classify motor state as ON or OFF using participants' medication diaries as ground truth. The CNN comprised convolutional layers that extracted spatial features from each a_{mag} window, followed by fully connected layers that learned the relationship between extracted features and motor state. Feature-wise Linear Modulation (FiLM) was applied to downweight windows with higher uncertainty by reweighting extracted features. A separate system was trained for each participant using 4-fold cross-validation, with folds constructed using a chronological splitting scheme to prevent information leakage. Mean absolute error (MAE) quantified forecasting accuracy, and balanced accuracy quantified classification performance.

To confirm the impact of uncertainty modulation on system performance, FiLM-based uncertainty conditioning was ablated. Balanced accuracy was compared between the full and the ablated model within each participant. Group-level differences were assessed using permutation testing.

Mechanisms explaining performance changes under *uncertainty awareness* were examined in three steps for each participant. First, each window was labeled as predominantly *epistemic* or *aleatoric* by computing a signal-to-noise ratio (SNR, 12 Hz). Windows with positive SNR indicated uncertainty arising from limited system knowledge, and were labeled as *epistemic*. Windows with negative SNR indicated dominance of noise and were labeled as *aleatoric*. Second, to evaluate the role of *epistemic* windows on overall system uncertainty, median CI width was compared between *epistemic* and *aleatoric* windows with the Mann–Whitney U test. Windows were treated as independent observations. Third, tracing gradient descent (TracIn) was applied to the uncertainty-aware CNN to quantify the influence of individual windows on prediction outcomes. For each prediction, the median TracIn influence score of *epistemic* windows was compared with that of *aleatoric* windows using the Wilcoxon signed-rank test, assuming paired observations within the same prediction. This assessed whether performance gains under *uncertainty awareness* were associated with reduced influence of *epistemic* compared with *aleatoric* uncertainty (α set at 0.05).

III. RESULTS

Forecasting accuracy was high for all participants, with an MAE below 0.1 m/s². The uncertainty-aware CNN accurately classified ON and OFF states for all participants, achieving a group-level balanced accuracy of 0.74 (Fig. 1(a)). Ablating FiLM-based uncertainty modulation reduced balanced accuracy by 11% ($p < 0.001$, Fig. 1(a)). This reduction varied across participants and was greater in participants with more severe motor symptoms, greater gait dysfunction, or stronger motor response to medication (P2 and P3) than in those with milder impairment (P1 and P4). Across all participants, overall system uncertainty was predominantly *epistemic* rather than *aleatoric* ($p < 0.001$). Performance gains under *uncertainty awareness* coincided with moderately-to-strongly lower influence of *epistemic* relative to *aleatoric* windows ($p < 0.01$, Fig. 1(b)).

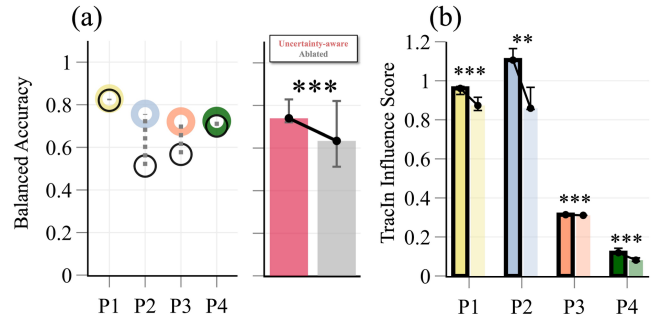


Fig. 1. System performance. (a) Participant-level mean balanced accuracy under *uncertainty awareness* (colored circles) and after ablation (black circles) (left), and group-level median balanced accuracy (right). (b) Participant-level median influence (unitless) for *aleatoric* (bars with borders) and *epistemic* (bars without borders) windows under *uncertainty awareness*. Error bars represent $\pm 95\%$ CI of the median. *** $p < 0.001$, ** $p < 0.01$ (1000 permutations).

IV. DISCUSSION

Our results show that performance gains under uncertainty primarily reflect reductions in *epistemic* rather than *aleatoric* uncertainty. This interpretation was further supported by greater performance gains under uncertainty modulation in participants with more difficult behavior to learn. This suggests that future wearable sensing systems for PD should explicitly

quantify *epistemic* uncertainty to identify system knowledge gaps. Such information could guide adaptive data collection strategies in which additional data with appropriate contextual labels (e.g., environmental context) are acquired during periods of high uncertainty. These labeled data could help disentangle the factors driving system knowledge gaps to better reduce epistemic uncertainty and improve system reliability. Validation in larger datasets and other relevant applications in PD is also required for generalizability.

V. CONCLUSION

This proof-of-concept study identifies *epistemic* uncertainty as a primary mechanism driving performance gains in uncertainty-aware wearable sensing for PD.

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